

Ritam Chakraborty

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EDUCATION

University of California, Berkeley

GPA: 3.87 • Berkeley, CA

Bachelor of Arts in Computer Science, Data Science

Expected Graduation: 05/2028

- Relevant Coursework: Data Structures, Algorithms, Probability and Random Processes, Discrete Mathematics and Probability Theory, Data Science, Numerical Analysis, Linear Algebra, Multivariable Calculus

EXPERIENCE

Software Engineering Intern

Richmond, VA

Capital One

06/2026 – Present

- Designed and delivered an automated recovery framework using AWS EventBridge, SQS with DLQ, and Lambda to resolve recurring receipt drops caused by transient partner API outages, achieving an 88% reduction in manual on-call interventions
- Shipped an idempotent retry architecture incorporating check-before-write state validation and micro-batching to systematically prevent duplicate receipt uploads and Lambda timeout failures, ensuring reliable high-volume processing at scale
- Produced comprehensive end-to-end QA integration tests covering retry success paths, DLQ routing, silent drops, and duplicate record handling to validate system correctness and ensure full production readiness across all critical failure modes

Monetization & Analytics Lead

Berkeley, CA

Berkeleytime

09/2025 – Present

- Leading a team of 5 engineers to ship new features and gather analytics for Berkeleytime, a course discovery platform for UC Berkeley serving over 10,000 students, building backend services with Express.js, MongoDB, Redis, and GraphQL
- Deployed a threshold-based course alert system that monitors enrollment changes and triggers email notifications, integrating background jobs and parallel workers, achieving sub-minute alert latency and reducing missed notifications by over 80%
- Developed a user activity scoring system with configurable decay formulas incorporating sign-in recency, session time, and feature usage, alongside dashboard visualizations enabling the team to prioritize development based on real usage patterns

Full Stack Engineering Lead (Contract)

Remote

Impact Observatory

02/2026 – 04/2026

- Managed a team of 6 engineers for the development of a geospatial analytics platform for Impact Observatory, architecting a React/TypeScript frontend, Python FastAPI backend, and LangGraph-based LLM orchestration layer integrated with OpenAI
- Designed an LLM agent tooling pipeline using LangGraph with RAG over 100+ product documents for intent routing
- Built a natural language query layer that converts user queries into GeoJSON boundaries for geospatial search and filtering

Software Engineering Intern

Berkeley, CA

Someldea AI

03/2025 – 12/2025

- Spearheaded development of an NLP-powered transcript analysis platform using Python with FastAPI, condensing 100+ pages of financial documents into actionable summaries and reducing manual research time for 50+ paid users by over 90%
- Migrated 1,000+ documents from Google Cloud Firestore to Amazon RDS, designing normalized relational schemas and building Python/SQL ETL pipelines, enabling indexed queries and reducing data retrieval time by 60% under increased load

PROJECTS

vigil • Rust, Tokio

- Built a Rust log analytics engine backed by an async Tokio daemon that ingests JSON event streams and maintains separate in-memory field and time indexes, serving sub-millisecond filtering and aggregation to concurrent clients over a Unix socket
- Supported latency and error-rate analysis across millions of events through a custom query language and hand-written parser with boolean predicates, time-range filters, and streaming aggregations including counts, averages, and p50/p99 percentiles
- Reduced daemon restart time by 1.6-1.8 times (up to 840 ms at 1M events) with a checksummed write-ahead log and periodic snapshots, so recovery replays only records newer than the last snapshot instead of re-parsing the entire log from disk
- Validated recovery gains with a 600-run statistical benchmark suite (Welch's t-tests, 95% confidence intervals) across graceful and crash restarts, uncovering and fixing two durability bugs that silently blocked snapshot recovery above 47K events

pgraph • Go

- Created a probabilistic graph inference engine in Go that models directed graphs with independent Bernoulli edge probabilities, enabling exact and sampled reachability analysis and probabilistic path queries for risk and reliability use cases
- Optimized core probabilistic graph algorithms, including max-likelihood path search via a modified Dijkstra's algorithm, top-K path enumeration with Yen's algorithm, and concurrent Monte Carlo reachability simulation using goroutines
- Designed a custom query DSL with JSON graph and result serialization for expressive, composable probabilistic queries

SKILLS

Languages: Rust, Python, Go, C++, Java, SQL, TypeScript

Backend: FastAPI, Flask, Spring, Node.js, Express.js

Systems & Infrastructure: Docker, Kubernetes, AWS, Linux, Git

Data & ML: Pandas, NumPy, SciPy, Matplotlib, PyTorch, TensorFlow

Databases: PostgreSQL, MongoDB, DynamoDB